Robust Document Classification Under Shift

Abstract. We study the robustness of transformer-based document classifiers under distribution shift. Using three public benchmarks, we show that a simple data augmentation strategy improves out-of-domain accuracy by 4.2 points on average.

1. Introduction

Large language models are increasingly used to triage scientific submissions and screen documents. Prior work has examined adversarial robustness at the token level, but comparatively little attention has been paid to document-level perturbations.
In this paper we propose a lightweight augmentation pipeline and evaluate it across news, legal, and biomedical corpora.

2. Method

We apply paraphrase, section-shuffle, and formatting noise to each training document and fine-tune a base encoder for three epochs with a learning rate of 2e-5.